National Technical University of Athens
School of Electrical and Computer Engineering
Division of Computer Science

Continuous Machine Learning for Cooperative, Connected and Automated Mobility applications

DIPLOMA THESIS

of

Georgios A. Kyriakopoulos

Supervisor: Panayiotis Tsanakas
Professor

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Committee

Copyright

Περίληψη

Γράψτε εδώ την ελληνική περίληψη της διπλωματικής σας εργασίας.

Περιγράψτε συνοπτικά:

- Το πρόβλημα που αντιμετωπίζετε
- Τη λύση που προτείνετε
- Τα κύρια αποτελέσματα
- Τη συνεισφορά της εργασίας

Λέξεις Κλειδιά: Λέξη κλειδί 1, Λέξη κλειδί 2, Λέξη κλειδί 3, Λέξη κλειδί 4

Abstract

Write your English abstract here.

Briefly describe:

- The problem you are addressing
- Your proposed solution
- Main results and contributions
- Impact and significance

Keywords: Keyword 1, Keyword 2, Keyword 3, Keyword 4

Acknowledgements

I would like to thank...

[Write your acknowledgements here. Usually thank your supervisor, committee, collaborators, friends, and family.]

Contents

List of Figures

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Extended Abstract in Greek

Εισαγωγή

Γράψτε μια σύντομη εισαγωγή στο πρόβλημα...

Θεωρητικό Υπόβαθρο

Περιγράψτε το θεωρητικό υπόβαθρο...

Μεθοδολογία

Περιγράψτε την προσέγγισή σας...

Υλοποίηση

Περιγράψτε την υλοποίηση...

Αποτελέσματα

Παρουσιάστε τα αποτελέσματα...

Συμπεράσματα

Συνοψίστε τη συνεισφορά σας...

Chapter 1

Introduction

1.1 Motivation

Why is this problem important?

1.2 Problem Statement

What problem are you solving?

1.3 Contributions

What did you contribute?

- Contribution 1
- Contribution 2
- Contribution 3

1.4 Thesis Organization

Brief overview of chapters.

[1]

Chapter 2

Background and Related Work

2.1 Background

Provide necessary background.

2.2 Related Work

Review related work.

2.3 Summary

Summarize and position your work.

Chapter 3

Methodology

3.1 Overview

High-level overview of your approach.

3.2 Dataset Generation

How you generated data with SUMO, etc.

3.3 Model Architecture

Your ML model design.

3.4 Drift Detection

Concept drift detection approach.

Chapter 4

Implementation

4.1 Platform Architecture

System design and components.

4.2 Technology Stack

Tools, frameworks, languages used.

4.3 Implementation Details

Key implementation decisions and challenges.

Chapter 5

Evaluation and Results

5.1 Experimental Setup

Datasets, metrics, baselines.

5.2 Results

Present your experimental results.

5.3 Discussion

Analyze and interpret the results.

Chapter 6

Conclusion and Future Work

6.1 Summary

Recap your contributions.

6.2 Limitations

Honest assessment of limitations.

6.3 Future Work

What could be done next?

Bibliography

[1] Doe, J. and Smith, J. “An Example Paper”. In: *Journal of Examples* 1.1 (2024).